



## Interval of XOR (intervalxor)

Mr. X enjoys solving and especially talking a lot about bitwise operations, even at the New Year's Day party. While celebrating with his friends, he came up with the following challenge.

There are  $q$  queries, and in each query you are given a positive integer  $a_i$  ( $i = 0 \dots q - 1$ ). You are required to answer the following question:

What is the bitwise XOR of the integers in the interval  $[0, a_i]$  and what is the maximum bitwise XOR we can get by removing **exactly** one integer from that interval?

Your task is to solve the challenge of Mr. X.

 Among the attachments of this task you may find a template file `intervalxor.*` with a sample incomplete implementation.

### Input

The first line of the input contains  $q$ , the number of queries.

Each of the next  $q$  lines contains a positive integer  $a_i$ , describing a query.

### Output

You need to write  $q$  lines, the answers to each query. Every line should contain two numbers, namely, the bitwise XOR of the integers in the interval  $[0, a_i]$ , and the maximum bitwise XOR we can get by removing one integer from the interval.

### Constraints

- $1 \leq q \leq 200\,000$ .
- $1 \leq a_i \leq 1\,000\,000\,000$  for each  $i = 0 \dots q - 1$ .

### Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- |                                                                                     |                              |
|-------------------------------------------------------------------------------------|------------------------------|
| – Subtask 1 (0 points)                                                              | Examples.                    |
|  |                              |
| – Subtask 2 (25 points)                                                             | $q, a_i \leq 1000$ .         |
|  |                              |
| – Subtask 3 (20 points)                                                             | $a_i \leq 10000$ .           |
|  |                              |
| – Subtask 4 (25 points)                                                             | $a_i \leq 10^6, q \leq 50$ . |
|  |                              |
| – Subtask 5 (30 points)                                                             | No additional limitations.   |
|  |                              |

# Examples

| input | output |
|-------|--------|
| 5     | 1 29   |
| 29    | 15 15  |
| 14    | 1 9    |
| 9     | 12 15  |
| 12    | 0 31   |
| 31    |        |